MLFF MLCV-SELECT1 run-local full checkpoint selection specification

Status: implemented in mdstats 0.20.135a0; replay-degradation semantics corrected in mdstats 0.20.140a0 (`MLCV-SELECT1`).

Scope

MLCV-SELECT1 consumes each run's current MLCV-RANK1 top-K shortlist and performs authoritative run-local full-validation checkpoint acceptance. It never pools checkpoints across folds/seeds, never uses an outer CV fold to select an epoch, and does not choose the production model.

Inputs

Each completed run provides immutable current RANK1 evidence (default top five), the target/replay score weights and target criterion, the retained safety-metric policy, the run-correct complete target selection artifact, complete independent TRUE_DFT replay validation `R_full`, and authenticated candidate/foundation lineage.

For a CV fold, target selection is `V_i_full` with `TARGET_CHECKPOINT_SELECTION` authority. Outer fold `C_i` has only `TARGET_OUTER_CV_EVALUATION` authority and is forbidden. For a final-development run, target selection is held-out `D_full` with `TARGET_FINAL_VALIDATION` authority.

Matched replay baseline

SELECT1 uses the foundation baseline measured on the exact same complete replay domain:

`R0_full = RMSE(foundation, R_full).`

For every candidate:

`DeltaR_full = R_full - R0_full.`

Both raw `R_full` and signed `DeltaR_full` are persisted. Negative degradation is valid and beneficial. Baseline model SHA-256 plus replay artifact digest/SHA bind the zero point; an `R_light` baseline cannot substitute for `R0_full`.

Full acceptance gates

Every retained RANK1 checkpoint is fully evaluated; SELECT1 does not stop after the first survivor. The component-wise gates are

`T_full <= T_max`

and

`DeltaR_full <= DeltaR_max,`

where, unless explicitly overridden,

`DeltaR_max = (w_T / w_R) * T_max.`

For diagnostics only, the equivalent absolute replay ceiling is

`R_full <= R0_full + DeltaR_max.`

Thus with `R0_full = 75.281 meV/A`, `T_max = 30 meV/A`, and 1:1 weights, a candidate replay RMSE up to `105.281 meV/A` can pass the replay-retention gate. A raw replay value above 30 meV/A is not itself a failure.

Configured energy MAE, focus-force RMSE, stress RMSE, and worst-condition force RMSE limits remain hard gates where present. No scalar score can compensate for a failed target, replay-degradation, or retained safety gate.

`target_stop_fraction` and `replay_stop_multiplier` are STOP1 training-control factors only and do not modify SELECT1 acceptance.

Representative score

Only full-gate survivors receive

```
S_full = (w_T*T_full + w_R*DeltaR_full)/(w_T+w_R).
```

Deterministic ties prefer lower score, lower target RMSE, lower replay degradation, lower absolute replay RMSE, better original lightweight rank, earlier epoch, then checkpoint SHA-256.

Durable result

Each run receives one immutable current run-selection record with either `representative_selected` or explicit `no_representative`. Current candidate evidence binds target RMSE, absolute replay RMSE, `R0_full`, signed replay degradation, degradation budget, diagnostic absolute ceiling, baseline model SHA, exact full-domain lineage, and score.

Historical v1 records keep their original raw-absolute replay meaning and digest. They are never treated as current degradation evidence without recomputation against authenticated matching foundation baselines.

Gate boundary

MLCV-SELECT1 does not evaluate a fold representative on untouched `C_i`, aggregate CV statistics, compare final-development seeds, export a production/committee model, perform physical NVE verification, or activate locked target test `E`. Those responsibilities remain AGG1, FINAL1, and VERIFY1.